

HW3 All of the following exercises are from LADW.

6.1. True or false (explain your answers):

- a) Any system of linear equations has at least one solution;
- b) Any system of linear equations has at most one solution;
- c) Any homogeneous system of linear equations has at least one solution;
- d) Any system of n linear equations in n unknowns has at least one solution;
- e) Any system of n linear equations in n unknowns has at most one solution;
- f) If the homogeneous system corresponding to a given system of a linear equations has a solution, then the given system has a solution;
- g) If the coefficient matrix of a homogeneous system of n linear equations in n unknowns is invertible, then the system has no non-zero solution;
- h) The solution set of any system of m equations in n unknowns is a subspace in $\mathbb{R}^n$;
- i) The solution set of any homogeneous system of m equations in n unknowns is a subspace in $\mathbb{R}^n$.

7.1. True or false (explain your answers):

- a) The rank of a matrix is equal to the number of its non-zero columns;
- b) The $m \times n$ zero matrix is the only $m \times n$ matrix having rank 0;
- c) Elementary row operations preserve rank;
- d) Elementary column operations do not necessarily preserve rank;
- e) The rank of a matrix is equal to the maximum number of linearly independent columns in the matrix;
- f) The rank of a matrix is equal to the maximum number of linearly independent rows in the matrix;
- g) The rank of an $n \times n$ matrix is at most n ;
- h) An $n \times n$ matrix having rank n is invertible.